sPHENIX Self-Paced Course

Syllabus and Study Plan

Prepared for P. D. Lewis (Ohio University) November 2026

Built on the Summer-2026 sPHENIX School and Workfest (BNL, 27 July – 14 August 2026), public sPHENIX papers and your own notes

1 Purpose

This course sharpens your working knowledge of the whole sPHENIX experiment: the physics programme, the accelerator, every detector subsystem, the trigger and data acquisition, the Fun4All software and simulation, reconstruction and calibration, and the analysis workflow from good-run list to publication. You already know the EMCal π^0/η corner of the experiment well; the goal is to understand the parts around it well enough to read any sPHENIX analysis note, ask good questions in meetings, and make better decisions in your own Run-24 p+p π^0/η cross-section analysis.

How the course is built

Ten module PDFs, each 14–25 pages. Every module follows the same pattern: learning objectives and prerequisites; numbered steps, each with *goal*, *physics*, *how it is done in sPHENIX*, *checks* and *pitfalls*; "Connect to your analysis" boxes; suggested hands-on exercises on SDCC; self-test questions with answers; a glossary and references. The school talks behind each module are cited as [S##] (list in Section 7).

2 How to use this course

1. Work through one module per week (two weeks for Modules 6 and 10). Budget the hours in Table 1.
2. Each module is self-contained: it builds the material from first principles up to current sPHENIX practice, so you do not need the slides. The school talks it cites as [S##] are optional further reading (Indico event 32182, collaboration login).
3. Read the module once straight through, then a second time working the checks. The SDCC exercises are optional.
4. Answer the self-test questions *before* reading the answers. If you miss more than three, reread that step.
5. Keep a running log (one page per module) of what you learned and what you want to ask; bring it to group meetings.
6. After Module 10, write the capstone proposal and discuss it with your advisor.

3 Schedule

Dependencies

Modules 1–2 are prerequisites for everything. Modules 3–5 can be read in any order. Module 6 is needed before 7 and 8. Module 9 uses 2, 5, 6 and 7. Module 10 uses all modules.

Week	Module	Title	Hours
1	1	The sPHENIX physics program: QGP, jets, heavy flavour, cold QCD and spin	8
2	2	RHIC and the sPHENIX detector: layout, global detectors, runs and centrality	8
3	3	Calorimetry in sPHENIX: EMCal and HCal hardware	8
4	4	Tracking detectors: MVTX, INTT, TPC and TPOT	8
5	5	Trigger, DAQ and streaming readout	7
6–7	6	Fun4All framework, software stack and simulation	14
8	7	Calorimeter reconstruction and calibration: towers, clusters, photons, jets	10
9	8	Tracking reconstruction: seeding, fitting, alignment and TPC calibration	9
10	9	From data to a result: the sPHENIX analysis workflow	10
11–12	10	Case studies from the 2026 school and machine learning in sPHENIX	14
		Total	96

Table 1: Suggested 12-week plan (about 8–10 hours per week). The order follows the school: physics, detectors, software, reconstruction, analysis.

4 Modules

Module 1: The sPHENIX physics program: QGP, jets, heavy flavour, cold QCD and spin

File: `Module01_Physics_Program.pdf` **Week:** 1 **Time:** 8 h
Scope. Why sPHENIX exists; QGP and jet quenching theory basics; history from PHENIX to sPHENIX; the four topical groups (Bulk, Jet Structure, Heavy Flavor, Cold QCD) and their flagship measurements; spin physics with polarized protons; where neutral mesons fit.
You will be able to:

- Explain what the QGP is and why jets, heavy flavour and quarkonia probe it
- Describe the four topical groups and their flagship measurements
- Explain the cold-QCD and spin programme with polarized protons
- Place neutral-meson measurements (π^0, η, R_{AA}, cross sections) in the programme

School talks: [S01], [S02], [S03], [S04], [S05], [S06], [S07], [S44], [S50].
Checkpoint exercise: Read the two first Au+Au papers; write a one-page map of which sPHENIX measurement constrains which QGP property.

Module 2: RHIC and the sPHENIX detector: layout, global detectors, runs and centrality

File: `Module02_RHIC_and_Detector_Overview.pdf` **Week:** 2 **Time:** 8 h
Scope. RHIC accelerator and beam basics (bunches, crossing angle, luminosity, polarization); detector layout and coordinates; solenoid; global detectors MBD, ZDC, sEPD; vertexing; centrality and Glauber; event plane; collision systems and runs 2023-2025.
You will be able to:

- Describe RHIC beams: bunches, crossing angle, luminosity, polarization
- Draw the sPHENIX layout and coordinate system
- Explain MBD, ZDC and sEPD and how vertex, centrality and event plane are obtained
- Summarise the runs 2023–2025 and collision systems

School talks: [S01], [S08], [S48], [S49], [S05].
Checkpoint exercise: On SDCC, query the spin DB and DAQ DB for a run of your choice (crossing angle, scaledowns); plot the MBD vertex distribution for that run.

Module 3: Calorimetry in sPHENIX: EMCal and HCal hardware

File: `Module03_Calorimeters.pdf` **Week:** 3 **Time:** 8 h
Scope. Electromagnetic and hadronic showers; sampling calorimetry; EMCal tungsten-powder, scintillating-fiber blocks and SiPM readout; inner/outer HCal; towers, light guides, digitizers; energy resolution; test beams; calibration strategy overview (MIP, π^0, tower slope, cosmics).
You will be able to:

- Explain electromagnetic and hadronic shower development
- Describe the EMCal and HCal construction and SiPM readout
- Relate sampling fraction and stochastic/constant terms to resolution
- Outline the calibration chain from test beam to in-situ

School talks: [S09], [S08], [S47], [S42].
Checkpoint exercise: Draw the EMCal tower grid; estimate the shower containment for a 2 GeV photon; look up one tower's calibration in the CDB.

Module 4: Tracking detectors: MVTX, INTT, TPC and TPOT

File: `Module04_Tracking_Detectors.pdf` **Week:** 4 **Time:** 8 h
Scope. Charged-particle detection in silicon and gas; MVTX MAPS pixels; INTT strips and timing; TPC drift, gain (GEM), space charge; TPOT micromegas; integration times and pile-up; role of each in vertexing, momentum, timing.
You will be able to:

- Explain charge collection in silicon and gas detectors
- Describe MVTX, INTT, TPC and TPOT geometry and roles
- Explain integration time, timing and pile-up for each detector
- Relate detector design to momentum and DCA resolution

School talks: [S10], [S11], [S12], [S28].
Checkpoint exercise: Sketch the hits of a 1 GeV pion through MVTX, INTT and TPC; estimate the TPC drift time and what it implies for pile-up.

Module 5: Trigger, DAQ and streaming readout

File: `Module05_Trigger_DAQ_Streaming.pdf` **Week:** 5 **Time:** 7 h
Scope. GL1 and trigger bits, MBD and calorimeter (photon/jet) triggers, scaledowns/prescales and live time; DAQ architecture and event building; triggered vs streaming readout; from raw data (PRDF) to DSTs; production, tags and the FileCatalog; SDCC computing access (ssh, web).
You will be able to:

- Explain GL1 trigger bits, scaledowns and live time
- Describe MBD and calorimeter-based triggers
- Contrast triggered and streaming readout
- Follow data from PRDF to DSTs and find files in the FileCatalog

School talks: [S13], [S21].
Checkpoint exercise: Use the FileCatalog to list the DST segments of one run; read the GL1 scalers for that run.

Module 6: Fun4All framework, software stack and simulation

File: Module06_Fun4All_and_Simulation.pdf **Week:** 6–7 **Time:** 14 h
Scope. Fun4AllServer, SubsysReco lifecycle, node tree, input/output managers, macros, CDB and global tags, builds and installs (MYINSTALL), coresoftware/macros/analysis repos; Geant4 detector simulation (PHG4Reco), event generators (PYTHIA8, HIJING), single-particle guns and embedding, simulation productions; hands-on exercises.
You will be able to:

- Write and run a Fun4All macro with a custom SubsysReco
- Navigate the node tree and the CDB/global tags
- Build and install a module into a private area
- Run Geant4 simulation with generators, particle guns and embedding

School talks: [S14], [S13], [S47].
Checkpoint exercise: Write a minimal SubsysReco that prints cluster counts; run a single-π^0 gun through Geant4 and reconstruct it.

Module 7: Calorimeter reconstruction and calibration: towers, clusters, photons, jets

File: Module07_Calorimeter_Reconstruction.pdf **Week:** 8 **Time:** 10 h
Scope. Waveform fitting and TowerInfo; status and hot-tower maps; EMCal π^0 and HCal calibrations; clustering (template clustering, chi2/probability), topo clustering; photon identification and isolation; jet reconstruction (anti-kt, constituents, background subtraction, JES/JER).
You will be able to:

- Explain waveform fitting, TowerInfo and status maps
- Describe EMCal and HCal calibration methods
- Compare clustering algorithms and topo clusters
- Explain photon ID/isolation and jet reconstruction with background subtraction

School talks: [S15], [S16], [S24], [S17], [S18], [S42].
Checkpoint exercise: Run clustering on one DST segment with two cluster settings and compare cluster counts and energies (no physics claims).

Module 8: Tracking reconstruction: seeding, fitting, alignment and TPC calibration

File: Module08_Tracking_Reconstruction.pdf **Week:** 9 **Time:** 9 h
Scope. Clustering in silicon and TPC; seeding; ACTS Kalman fitting; vertexing; alignment (millepede-style); TPC distortions and calibration (lasers, space charge); track-calorimeter matching; performance evaluation (efficiency, momentum resolution, DCA).
You will be able to:

- Explain seeding and Kalman-filter fitting in ACTS
- Explain detector alignment
- Describe TPC distortions and their calibration
- Explain track–calorimeter matching and performance metrics

School talks: [S19], [S20], [S22], [S23], [S36], [S27].
Checkpoint exercise: Run the tracking QA macros on a small sample and identify each stage's output nodes.

Module 9: From data to a result: the sPHENIX analysis workflow

File: `Module09_Analysis_Workflow.pdf` **Week:** 10 **Time:** 10 h
Scope. Run QA and good-run lists; trigger selection and efficiency; luminosity and Vernier scans; centrality in practice; efficiency, acceptance and embedding; unfolding basics; systematic uncertainties (Barlow test); blinding/approval process (analysis notes, preliminaries, papers).
You will be able to:

- Build good-run lists and run QA
- Compute luminosity and understand Vernier scans
- Design efficiency/acceptance corrections and unfolding
- Estimate systematic uncertainties and navigate the approval process

School talks: [S43], [S33], [S41], [S26], [S25], [S05], [S34].
Checkpoint exercise: Write the full error-budget table for your π^0/η cross section and the run-selection flow chart.

Module 10: Case studies from the 2026 school and machine learning in sPHENIX

File: `Module10_Case_Studies_and_ML.pdf` **Week:** 11–12 **Time:** 14 h
Scope. Worked mini-case-studies from student flash talks (multiplicity, flow v_2, jets, isolated photons and spin asymmetries, quarkonia energy correlators, phi-hadron correlations, CME, UPC rho, Vernier scans, INTT efficiency, HCal calibration); ML in physics (classification, regression for momentum scale, NN photon ID); capstone project plan tied to his π^0/eta analysis.
You will be able to:

- Analyse the structure of real student analyses from the school
- Use ML for classification and regression in physics
- Plan a capstone project tied to your π^0/η analysis

School talks: [S30], [S31], [S32], [S34], [S35], [S40], [S37], [S38], [S39], [S29], [S27], [S25], [S26], [S41].
Checkpoint exercise: Write a two-page capstone proposal and present it to your advisor.

5 Capstone

Write a two-page proposal for one improvement to your π^0/η cross-section analysis that uses what you learned outside the EMCal (for example: a track-based isolation or conversion veto, an MBD-trigger bias study with the global detectors, a streaming-readout cross-check of luminosity, or an ML photon/π^0 discriminator for merged clusters). State the physics motivation, the data and simulation needed, the method, the expected gain, and how you would validate it.

6 Public readings

- sPHENIX Public Results: `https://www.sphenix.bnl.gov/PublicResults`
- sPHENIX Collab., charged hadron multiplicity in Au+Au, arXiv:2504.02240, JHEP 08 (2025) 075.
- sPHENIX Collab., transverse energy density in Au+Au, arXiv:2504.02242, PRC 112 (2025) 024908.
- sPHENIX Collab., dijet imbalance and acoplanarity in p+p, arXiv:2606.17184.

- sPHENIX Collab., isolated prompt photons in p+p, arXiv:2607.03892.
- INTT ladder and readout, NIM-A 1082 (2026) 171020, arXiv:2503.09105; INTT test beam, NIM-A 1086 (2026) 171312, arXiv:2509.00908.
- Aidala et al., sPHENIX EMCal and HCal prototypes, IEEE Trans. Nucl. Sci. (2018) 2901–2919, arXiv:1704.01461.
- PHENIX Collab., An Upgrade Proposal from the PHENIX Collaboration, arXiv:1501.06197.

7 School talks cited in this course

All talks are from the Summer-2026 sPHENIX School and Workfest, BNL (Indico event 32182; collaboration access). The modules summarise them in their own words; slides, plots and internal numbers are not reproduced.

Key	Title	Speaker(s)
[S01]	Welcome to sPHENIX school and sPHENIX Overview	J. Huang, M. Connors
[S02]	sPHENIX History	D. Morrison
[S03]	sPHENIX Theory Overview: exploring QGP dynamics with jets	Y. Mehtar-Tani
[S04]	Cold QCD TG Analysis Opportunities	G. Nukazuka, M. Sarsour
[S05]	Bulk TG Analysis Opportunities	M. Mitrankova, R. Belmont
[S06]	Heavy Flavor TG Analysis Opportunities	A. Frawley, C. Dean, G. Ottino, M. Rosati
[S07]	Jet Structure TG Analysis Opportunities	J. Putschke, Y. Go
[S08]	HCal and Global Detector Overview	J. Haggerty
[S09]	EMCal Overview	C. Woody
[S10]	MVTX Overview	C. Dean
[S11]	INTT Overview	A. Enokizono
[S12]	TPC and TPOT Overview	T. Hemmick
[S13]	DAQ Overview (with web-browser and ssh tutorials)	S. Liechty
[S14]	Fun4All Overview / sPHENIX computing tutorial	C. Pinkenburg
[S15]	EMCal Reco and Calibrations	B. Seidlitz
[S16]	HCal Reco and Calibrations	A. Menon
[S17]	Photon Reconstruction	Y. Go
[S18]	Jet Reconstruction	V. Bailey
[S19]	Tracking Reconstruction	J. Osborn
[S20]	Tracking Alignment	G. Ottino
[S21]	Streaming Analysis	M. Peters
[S22]	TPC Calibration	M. Mitrankova
[S23]	Track-Calo Matching	X. Yu
[S24]	Topo Clustering	A. C. Oliveira da Silva
[S25]	Charged hadron multiplicity in O+O using silicon seeds (flash)	H.-R. Jheng
[S26]	Charged-hadron multiplicity in O+O collisions (flash)	C.-W. Shih
[S27]	Machine Learning Momentum Scale Correction (flash)	A. Patton
[S28]	INTT hit detection efficiency study (flash)	T. Kikuchi
[S29]	J/ψ energy correlators in p+p (flash)	B. Sayki

Key	Title	Speaker(s)
[S30]	Transverse single-spin asymmetries of isolated prompt photons (flash)	J. Hwang
[S31]	Neural network discrimination of isolated photons (flash)	R. Koike
[S32]	What I have done and what I plan to do next (flash)	S.-Y. Chen
[S33]	Run QA for Silicon-Calo Tracking Analysis (flash)	M. Shigihara
[S34]	Measurement of v_2 in Au+Au (flash)	I. Omae
[S35]	First Step in Jet Analysis (flash)	R. Takaaze
[S36]	Tracking Performance Evaluation in sPHENIX (flash)	A. P. Dash
[S37]	Constraining the CME using SDM (flash)	J. Singh
[S38]	Angular correlations of ϕ mesons and high-pT hadrons (flash)	K. Gable
[S39]	UPC ρ^0 production in O+O and Au+Au, simulation (flash)	L. Mwibanda
[S40]	Joint jet mass and charge for quark/gluon jet separation (flash)	M. I. Abdulhamid
[S41]	Vernier Scan analysis of p+p and O+O in Run 25 (flash)	T. Harada
[S42]	HCal Calibration studies (flash)	A. M. Khan
[S43]	Jet Analysis	J. Park
[S44]	Cold QCD Lecture Series announcement	G. Nukazuka
[S45]	Closing	B. Sayki, J. Huang, M. Connors
[S46]	Workfest kickoff	S. Bathe, Y. Go
[S47]	Calorimeter test-beam simulation (workfest)	B. Seidlitz, J. Nagle, S. Bathe
[S48]	Spin dynamics at the RHIC accelerator complex (workfest lecture)	K. Hock
[S49]	Polarimetry at RHIC (workfest lecture)	X. Chu
[S50]	Cold-QCD lecture (workfest)	M. Cerutti